HYDROTECH

DATABASE ERD - CURRENT STRUCTURE

9 tables | 60 columns | 9 foreign keys | PPM: INT | Crow foot notation

30 September 2026

Profile
profile

KEY	COLUMN	DATA TYPE	NULL
PK	profile_id	INT	No
	section	ENUM [D]	No
	label	VARCHAR(100)	No
	value	TEXT	No
	sort_order	INT	No
FK	updated_by	INT	Yes

R1 · Last editor

Users
users

KEY	COLUMN	DATA TYPE	NULL
PK	user_id	INT	No
UQ	username	VARCHAR(50)	No
	password	VARCHAR(255)	No
	email	VARCHAR(100)	Yes
	role	ENUM [A]	No
	created_at	TIMESTAMP	Yes
	updated_at	TIMESTAMP	Yes

R3 · Schedule editor

Dithane Schedule
dithane_schedules

KEY	COLUMN	DATA TYPE	NULL
PK	schedule_id	INT	No
	interval_hours	INT	No
	duration_seconds	INT	No
	start_time	TIME	No
FK	updated_by	INT	Yes
	created_at	TIMESTAMP	No
	updated_at	TIMESTAMP	No

R2 · Last editor

LEGEND

PK = primary key; FK = foreign key; UQ = unique.
NULL: Yes means the column may be empty.
Two bars = exactly one; circle + bar = zero or one.
Circle + crow foot = zero to many (N).
Dashed line = non-identifying relationship.
R1-R9 refer to the relationships on page two.

Settings
settings

KEY	COLUMN	DATA TYPE	NULL
PK	setting_id	INT	No
UQ	type	ENUM [B]	No
	config_json	JSON	No
	updated_at	TIMESTAMP	Yes
FK	updated_by	INT	Yes

R4 · Monitoring category

R5 · History category

History Logs
history_logs

KEY	COLUMN	DATA TYPE	NULL
PK	history_id	INT	No
	day_name	VARCHAR(20)	No
	log_date	DATE	No
	log_time	VARCHAR(10)	No
FK	type	ENUM [B]	No
	status	VARCHAR(30)	No
	note	TEXT	Yes
	is_deleted	TINYINT(1)	No
	created_at	TIMESTAMP	Yes

Monitoring Logs
monitoring_logs

KEY	COLUMN	DATA TYPE	NULL
PK	monitoring_id	INT	No
FK	type	ENUM [B]	No
	day_name	VARCHAR(20)	No
	log_time	VARCHAR(10)	No
	value	VARCHAR(20)	No
	status	VARCHAR(30)	No
	note	TEXT	Yes
	is_deleted	TINYINT(1)	No
	created_at	TIMESTAMP	Yes
FK	ph_reading_id	INT	Yes
FK	ppm_reading_id	INT	Yes

R7 · PPM source

IMPORTANT RULES

Schedule: schedule_id = 1; at most one row.
pH: 0-14; PPM: non-negative integers.
In monitoring and notifications, the two reading IDs cannot both be set in the same row.
Both may be NULL for general records.

R6 · pH source

pH Readings
ph_readings

KEY	COLUMN	DATA TYPE	NULL
PK	ph_reading_id	INT	No
	ph_value	DECIMAL(6,4)	No
	recorded_at	TIMESTAMP	No

PPM Readings
ppm_readings

KEY	COLUMN	DATA TYPE	NULL
PK	ppm_reading_id	INT	No
	ppm_value	INT	No
	recorded_at	TIMESTAMP	No

R9 · PPM source

Notifications
notifications

KEY	COLUMN	DATA TYPE	NULL
PK	notification_id	INT	No
	label	VARCHAR(100)	No
	message	TEXT	No
	tone	ENUM [C]	No
	href	VARCHAR(255)	Yes
	is_read	TINYINT(1)	No
	created_at	TIMESTAMP	Yes
FK	ph_reading_id	INT	Yes
FK	ppm_reading_id	INT	Yes

R8 · pH notification source

ENUM VALUES

A: admin / user B: ppm / ph / dithane C: warning / success / info D: owner / team

The Dithane category remains in settings for log references; schedule parameters are stored in dithane_schedules.

The schedule relationship uses 0..1 because the entire schedule table allows at most one row.

Source: database/schema.sql. Column names and types match the current implementation.

HYDROTECH - ERD RELATIONSHIPS

Diagram reference - all foreign keys and cardinality rules

R1 profile.updated_by → users.user_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R2 settings.updated_by → users.user_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R3 dithane_schedules.updated_by → users.user_id

At most one schedule in the entire system; a user may be referenced by zero or one schedule.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R4 monitoring_logs.type → settings.type

One parent may be referenced by zero to many child rows.
Each child must reference exactly one parent (NOT NULL foreign key).

R5 history_logs.type → settings.type

One parent may be referenced by zero to many child rows.
Each child must reference exactly one parent (NOT NULL foreign key).

R6 monitoring_logs.ph_reading_id → ph_readings.ph_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R7 monitoring_logs.ppm_reading_id → ppm_readings.ppm_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R8 notifications.ph_reading_id → ph_readings.ph_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

R9 notifications.ppm_reading_id → ppm_readings.ppm_reading_id

One parent may be referenced by zero to many child rows.
A child may have no parent (NULL foreign key), or reference exactly one parent.

REFERENCE AND DELETION RULES

R1, R2, R3: deleting a user sets updated_by to NULL (ON DELETE SET NULL).

R4-R9: referenced parent rows cannot be deleted (ON DELETE RESTRICT).

settings.type is UNIQUE and can be referenced by a foreign key even though it is not the primary key.

monitoring_logs: ph_reading_id is only valid for type ph; ppm_reading_id is only valid for type ppm.

Monitoring and notification rows cannot reference both pH and PPM readings at once.

history_logs has no direct foreign key to readings or the Dithane schedule.

updated_by identifies only the last editor; it is not a complete audit trail.

ERD edges describe stored relationships; they do not establish a physical ESP32 connection.

Based on the project schema checked on 30 September 2026.